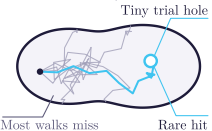
Finite-cavity differencing

Perturb the domain, then subtract

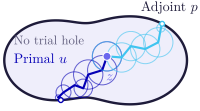


$$\frac{J(\Omega \setminus B_\varepsilon) - J(\Omega)}{\mathrm{cap}(B_\varepsilon)}$$

Tiny target; amplified noise

TWoS: score, check, insert

① Query the original domain



② Combine independent batches

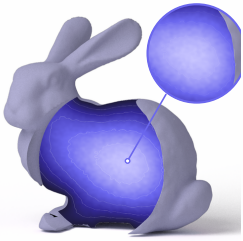
$$\widehat{D}_T = -\,\mathrm{avg}_{i \neq j}\, U_i P_j$$

③ Fresh check then insert



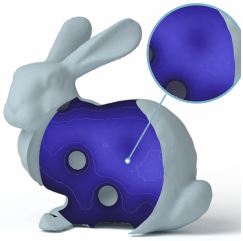
Initial temperature

Internal heating; cooled boundaries



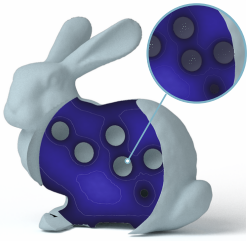
Random placement

6 cavities; 39.5% lower heat objective



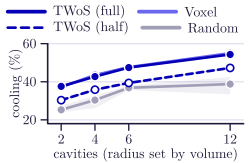
Voxel-adjoint placement

6 cavities; 48.1% lower heat objective



Similar cooling, half the volume

Full / half removal: 1.53% / 0.76%



64 sites: 1.39× faster; 81% less RAM

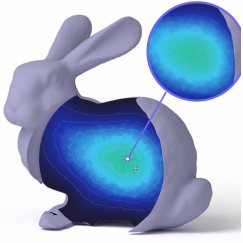
TD error: 5.9% (ours) vs 6.9% (voxel)

Separate query test; 16,384-walk reference



Where to insert

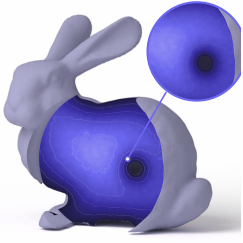
Predicted benefit $-D_T = u^2$





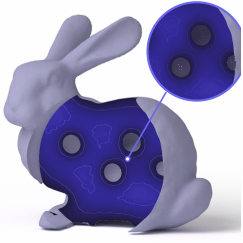
TWoS (1 cavity)

17.5% lower heat objective



TWoS (6 large cavities)

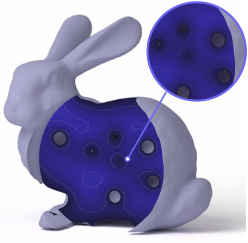
47.35% lower heat objective



Full budget: 1.529% removed; $r = 0.0500$

TWoS (12 small cavities)

47.07% lower heat objective



Half budget: 0.765% removed; $r = 0.0315$